propulsion mobility and contributes to a more efficient, multi-layered transportation framework in smart cities.

8.8 UAV Integration in Logistics and Distribution Systems

The integration of unmanned aerial vehicles (UAVs) in logistics and distribution systems represents one of the most significant advancements in modern supply chain transformation. Drones offer the ability to transport goods swiftly over short distances, bypassing ground-based limitations such as road congestion, difficult terrain, and limited transport infrastructure. Their use in last-mile delivery, warehouse coordination, real-time inventory monitoring, and emergency logistics has positioned UAVs as a strategic tool for enhancing the efficiency, responsiveness, and sustainability of supply chain operations. As global commerce increasingly prioritizes speed, reliability, and reduced environmental impact, UAVs are poised to play a central role in shaping future logistics ecosystems.

8.8.1 The evolution of last-mile delivery through drones

Last-mile delivery, the final leg in the distribution chain, has traditionally been the most time-consuming and costly part of logistics. Drones have introduced a transformative model where small-to-medium payloads can be transported directly from distribution centers to end-users or service nodes. E-commerce companies, healthcare logistics providers, and food delivery platforms have begun deploying UAV delivery fleets to shorten delivery times, particularly in congested urban settings or remote rural regions where traditional transportation is inefficient.

Early implementations focused on experimental and pilot projects, such as Amazon Prime Air and Google Wing trials, which validated UAV feasibility in delivering parcels weighing a few kilograms. Subsequent operational systems, such as Zipline in Africa, demonstrated the scalability